

Circuit Simulation Lab:7

Multiplexers

Introduction

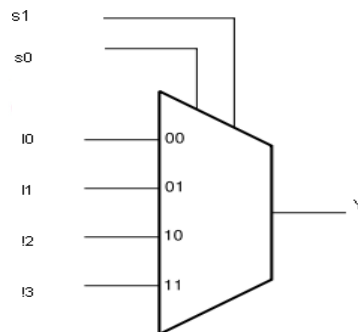
The purpose of this experiment is to introduce the design of simple combinational circuits, in this case multiplexers. A multiplexer is a digital circuit, which is used to select a single input from the multiple input lines.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

Logic Diagram



4:1 Multiplexer

s1	s0	Y
0	0	I0
0	1	I1
1	0	I2
1	1	I3

Truth Table

➡ Implement the boolean expression $F(A, B, C) = \sum m(2, 3, 6, 7)$ using a

multiplexer.

➡ Implement the boolean expression $F(A, B, C) = \sum m(0, 1, 3, 5, 7)$ using a multiplexer.

Describe the above in Verilog HDL and capture the Waveforms

- 1. Dataflow modeling*
- 2. Behavior modeling*
- 3. Structural modeling (Use the gates for designing the Mux)*
- 4. Design the above two problems using structural modeling – use Muxes as components*

